Where were the ports of the Periplus, and how did they work?

Vivek Shah

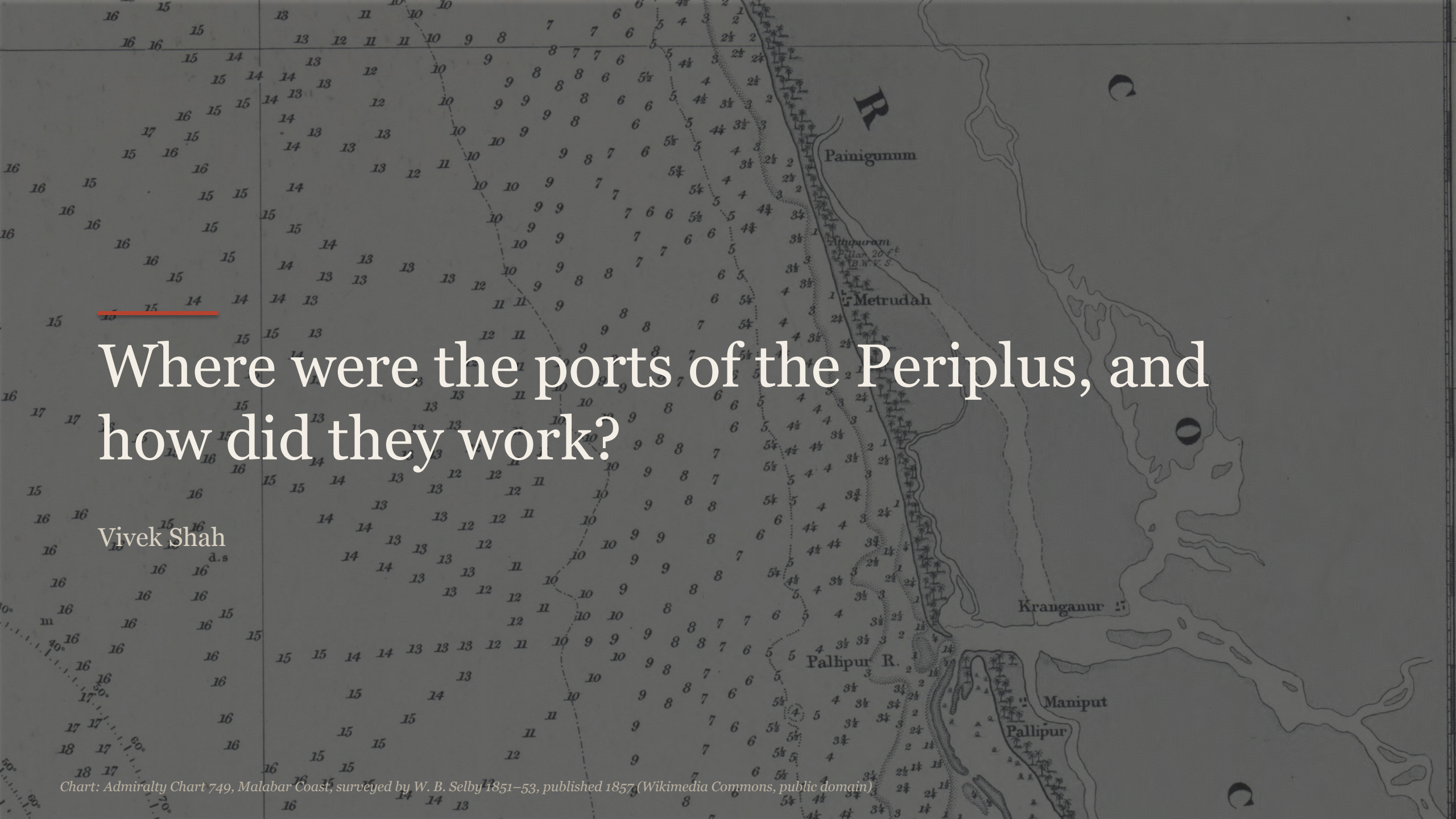
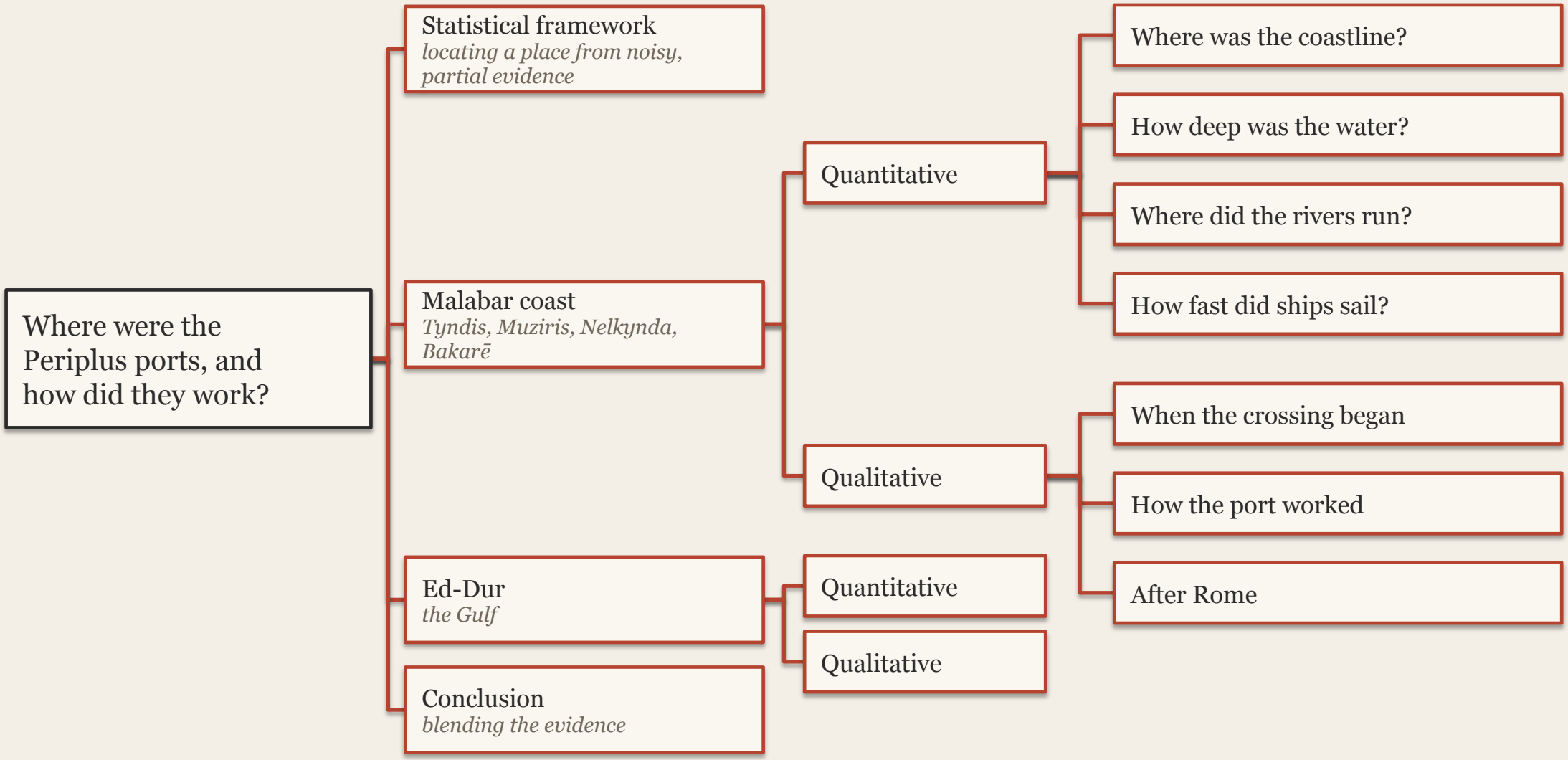


Chart: Admiralty Chart 749, Malabar Coast, surveyed by W. B. Selby 1851–53, published 1857 (Wikimedia Commons, public domain)

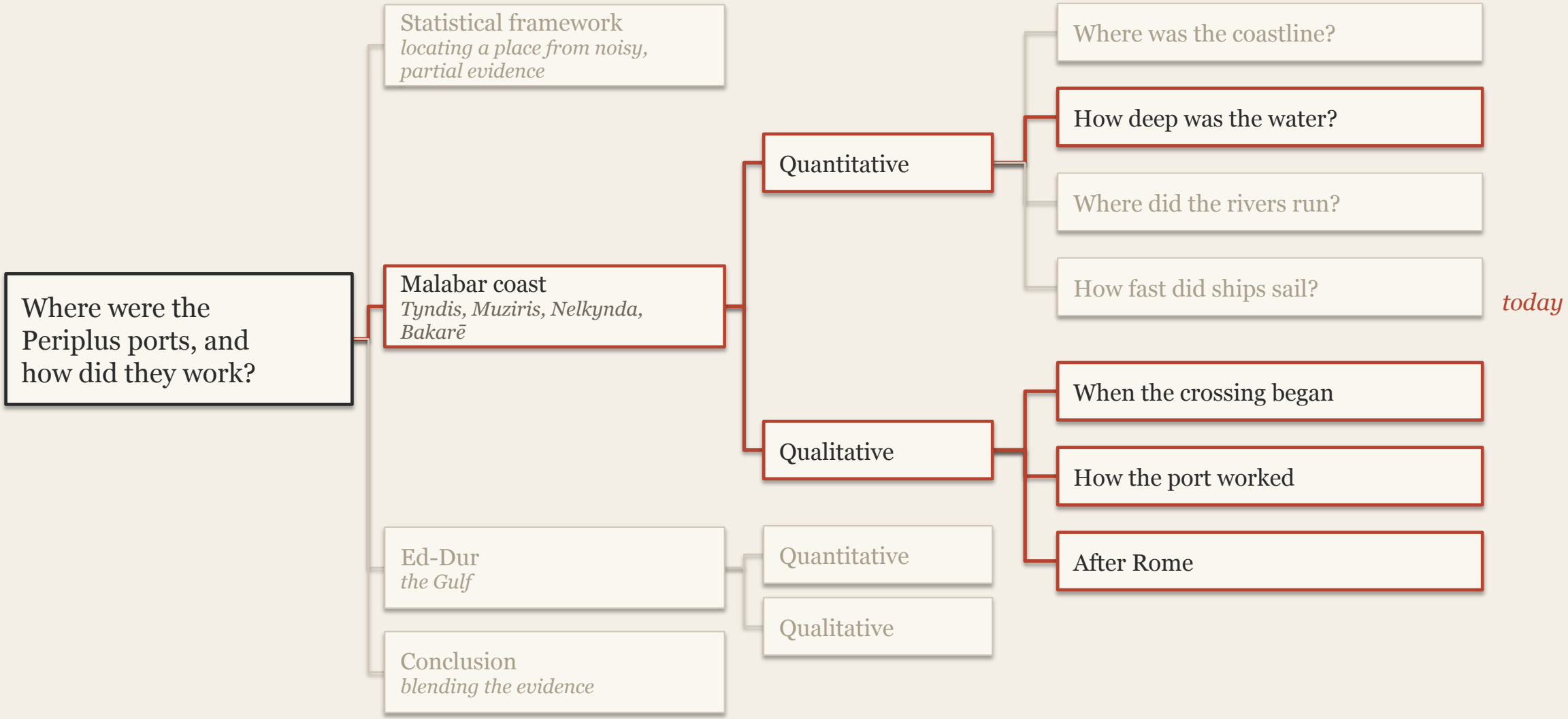
The thesis focuses on the Malabar coast and Ed-Dur, which were ports along the routes outlined in the Periplus

Map: PHGCOM, 'Map of the Periplus of the Erythraean Sea', Wikimedia Commons, CC BY-SA 4.0. Ed-Dur ring placed by the author

The thesis examines each port quantitatively to determine its location and qualitatively to understand how it functioned within the broader trading network



Today, we'll focus on Malabar Coast water depths and the ports' role in Indo-Roman trade.



Part I

How deep was the water at the mouths?

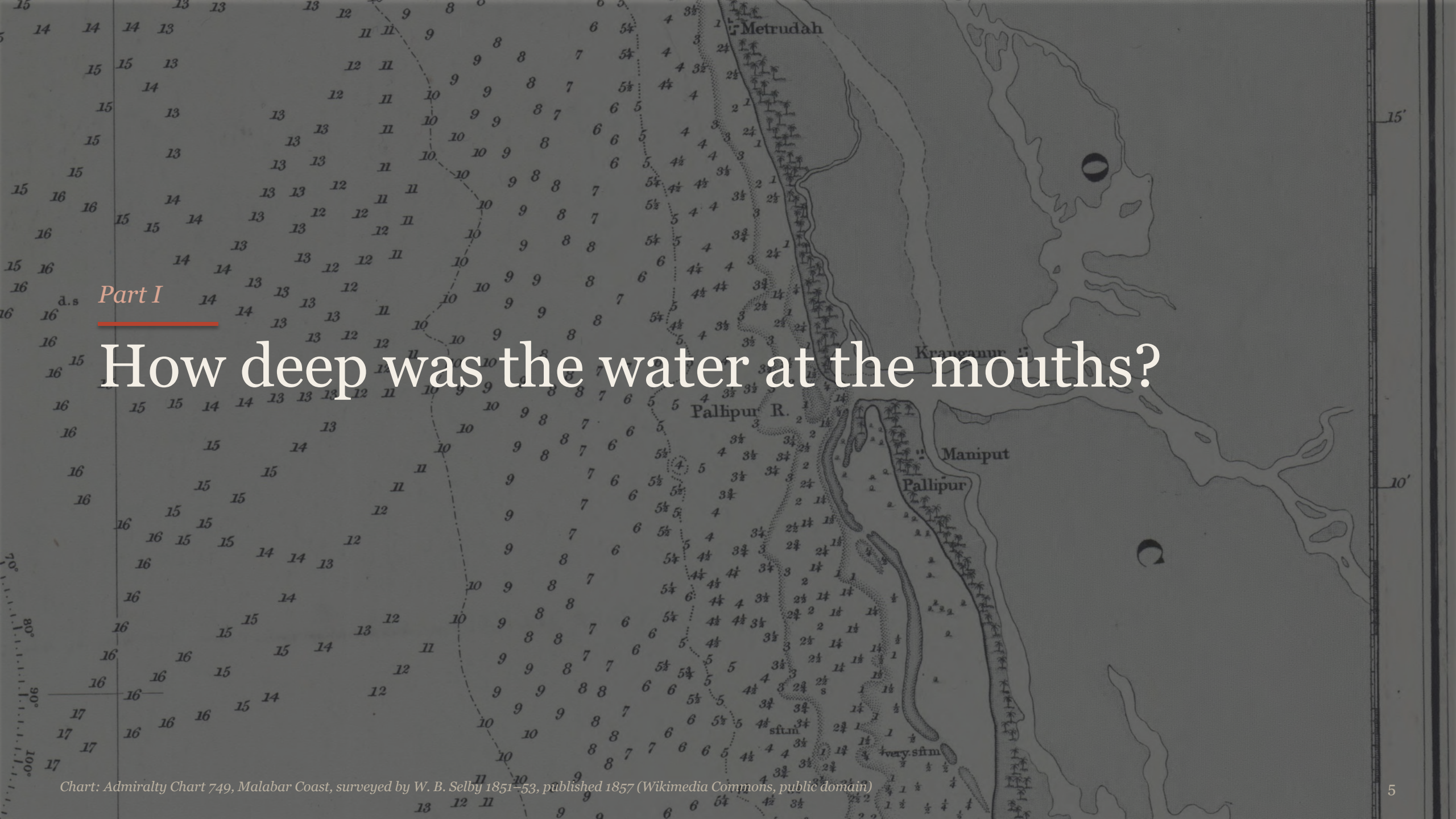


Chart: Admiralty Chart 749, Malabar Coast, surveyed by W. B. Selby 1851–53, published 1857 (Wikimedia Commons, public domain)

The primary evidence for coastal depths comes frrom British and Dutch admiralty surveys

Admiralty Chart 749, Malabar Coast, surveyed by W. B. Selby 1851–53, published 1857 (Wikimedia Commons, public domain)

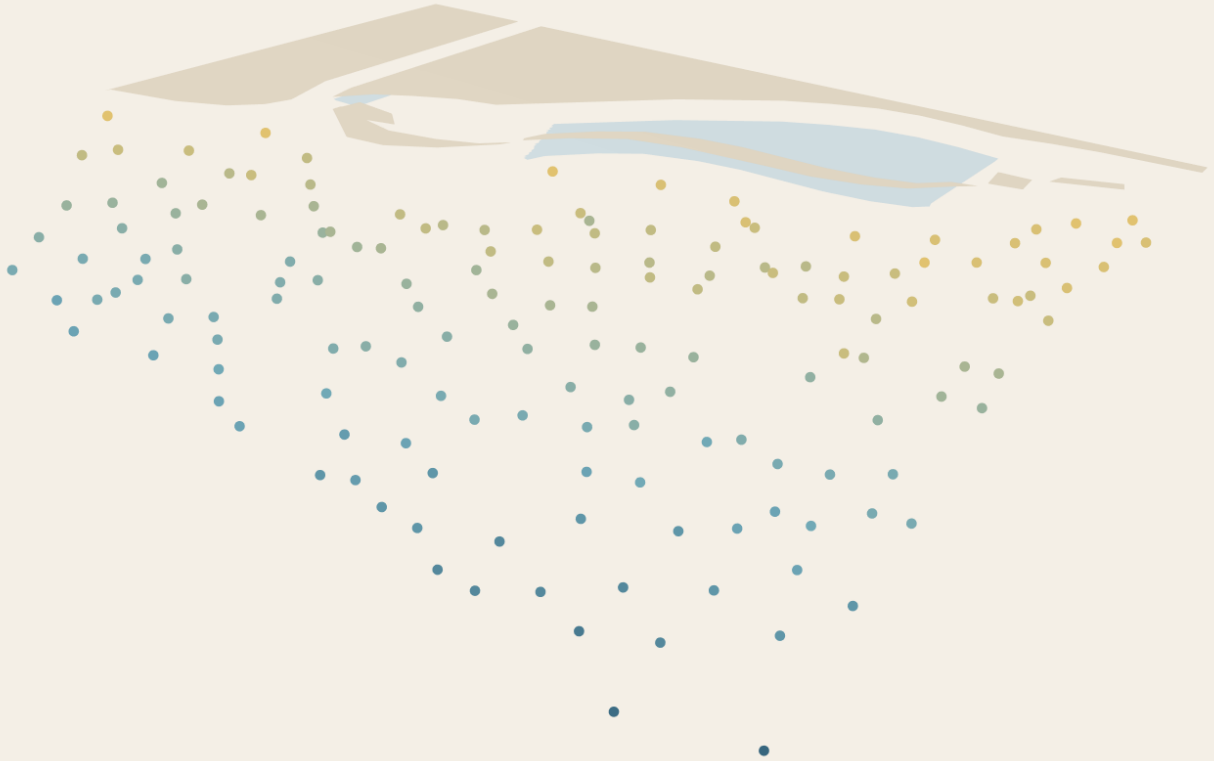
The numbers on this chart are depth (measured in fathoms), and these soundings are transcribed

Admiralty Chart 749, Malabar Coast, surveyed by W. B. Selby 1851–53, published 1857 (Wikimedia Commons, public domain)

The numbers on this chart are depth (measured in fathoms), and we transcribe these soundings

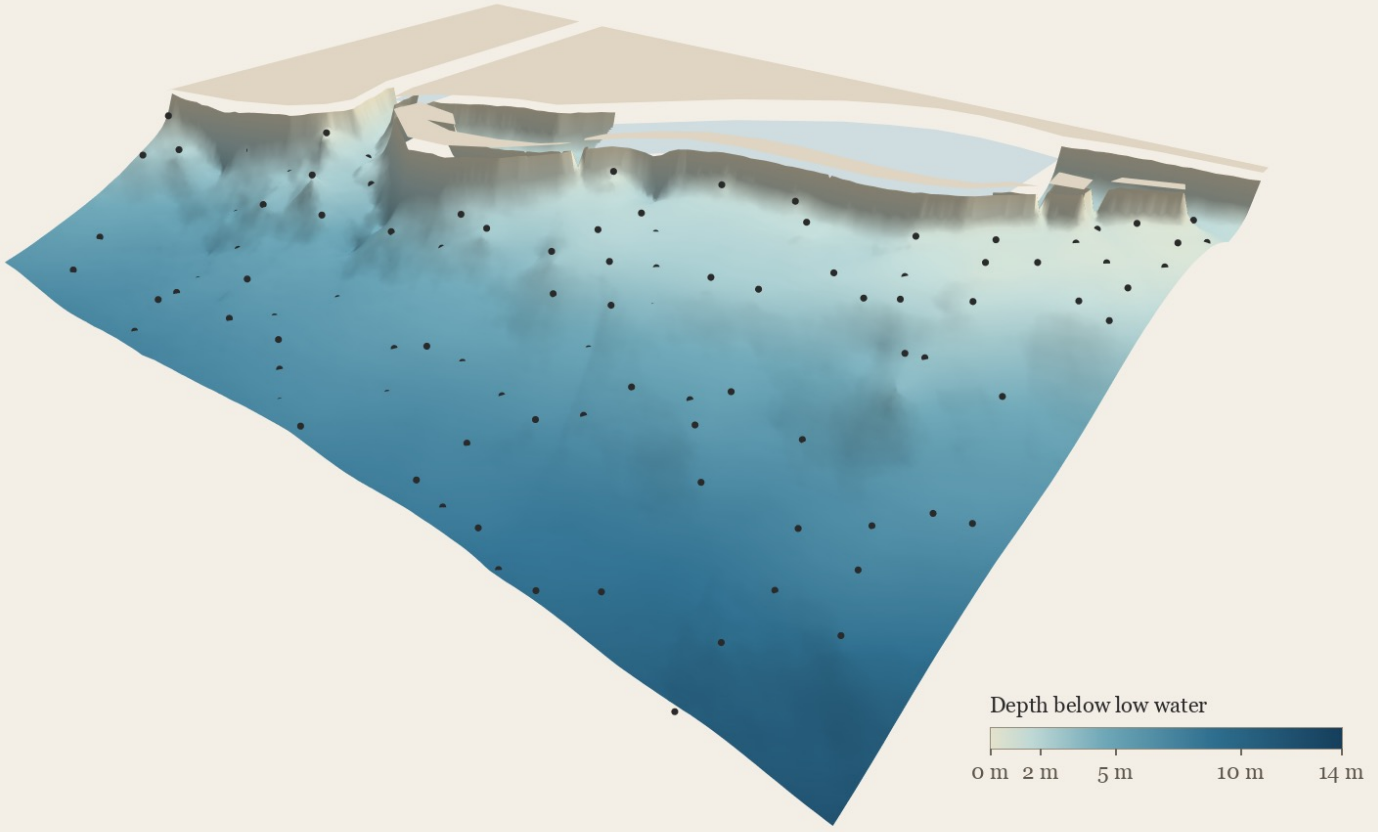
Admiralty Chart 749, Malabar Coast, surveyed by W. B. Selby 1851–53, published 1857 (Wikimedia Commons, public domain); points: the author's transcription

These depth soundings can then be used, along with the coordinates of these points, to map the coast in 3D



Soundings from Admiralty Chart 749; seabed modelled with a Gaussian process (R-INLA)

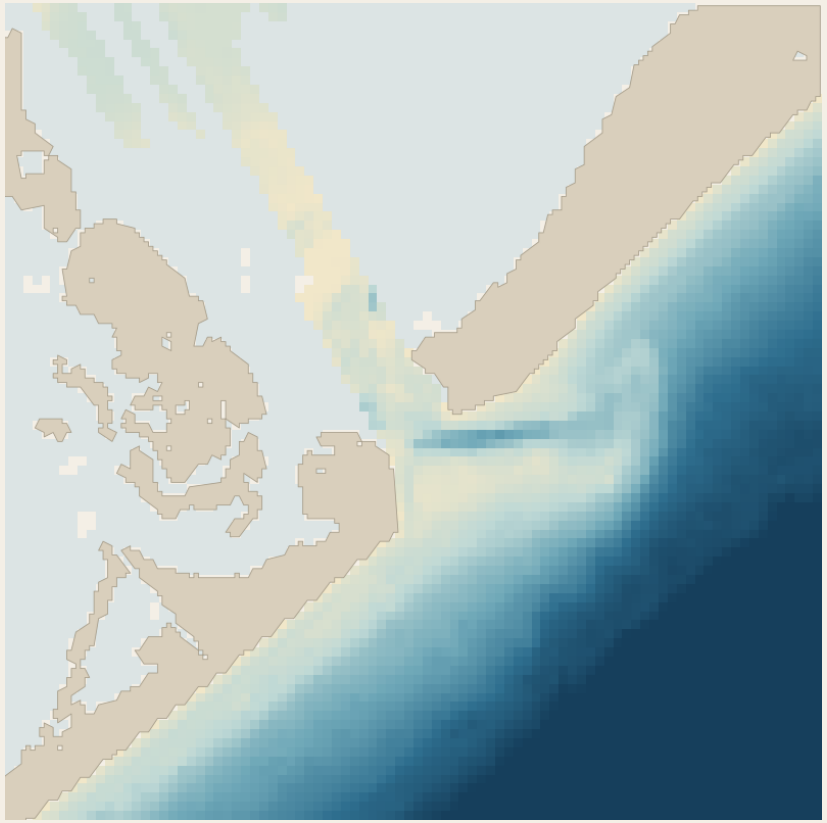
Established statistical models are used to fill in the depths between the points to get a richer sense of the coast



Soundings from Admiralty Chart 749; seabed modelled with a Gaussian process (R-INLA)

To test out the method, it can be used on a modern Texas inlet whose seabed is known

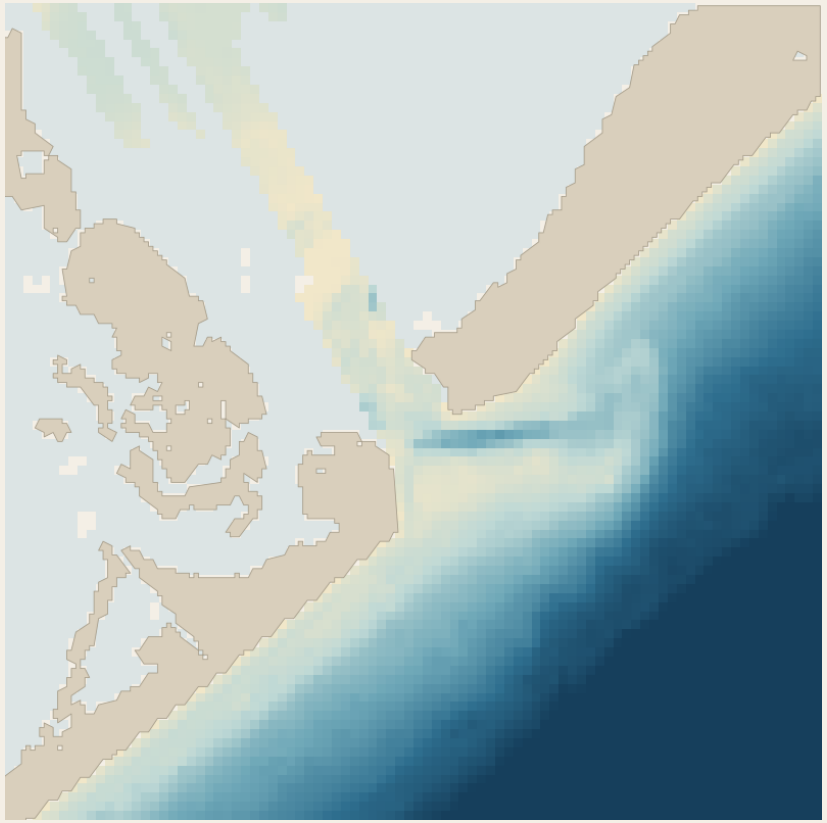
The real seabed



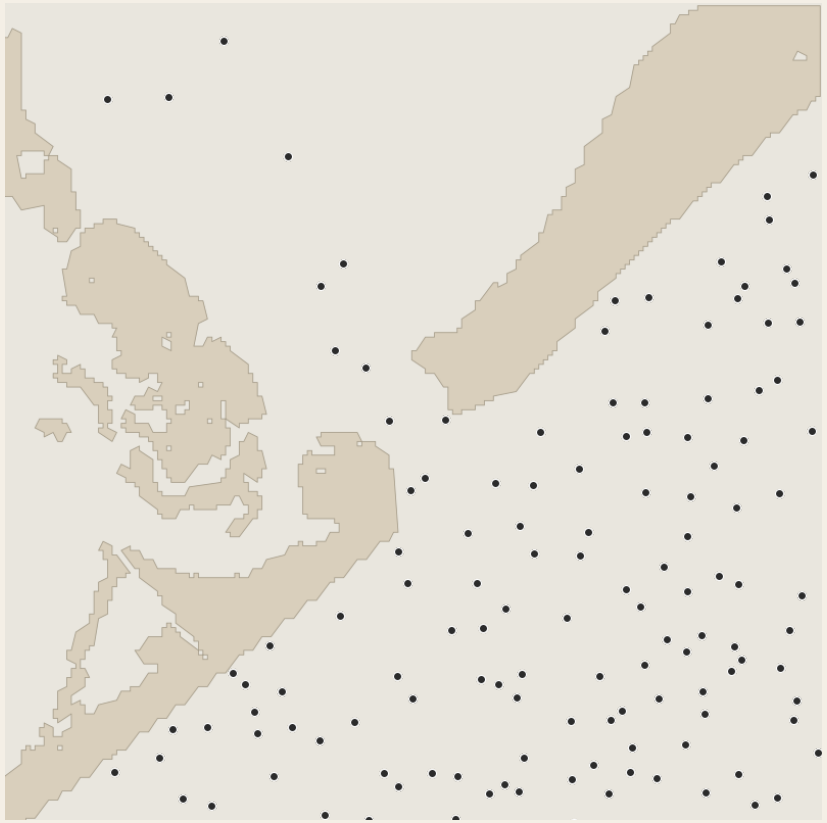
San Luis Pass, Texas: NOAA Continuously Updated DEM (CUDEM); survey simulated at the 1853 Kodungallur density

The seabed is sampled 130 times to create “soundings” that are similar to the ones in the British admiralty survey

The real seabed



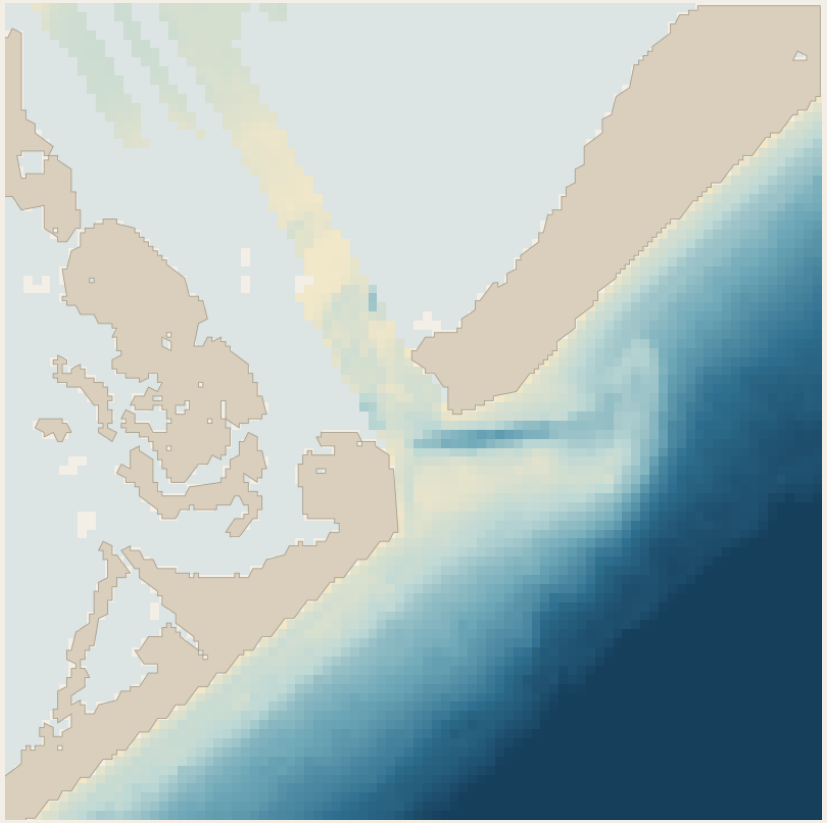
Rebuilt (130 soundings)



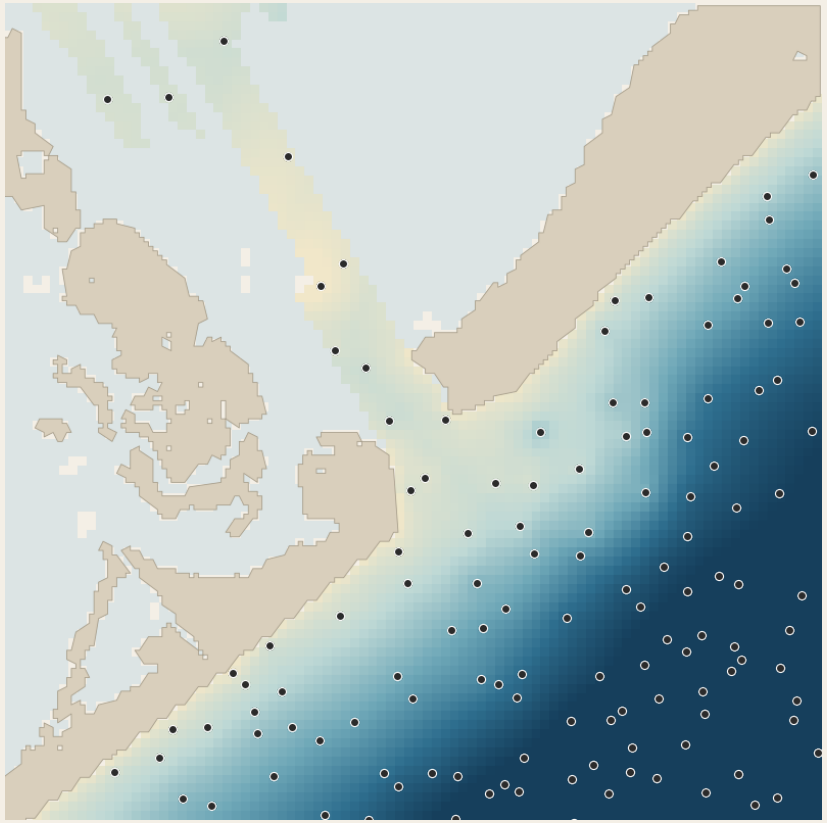
San Luis Pass, Texas: NOAA Continuously Updated DEM (CUDEM); survey simulated at the 1853 Kodungallur density

The seabed is then rebuilt from the soundings and the rebuilt version is extremely similar to the original

The real seabed



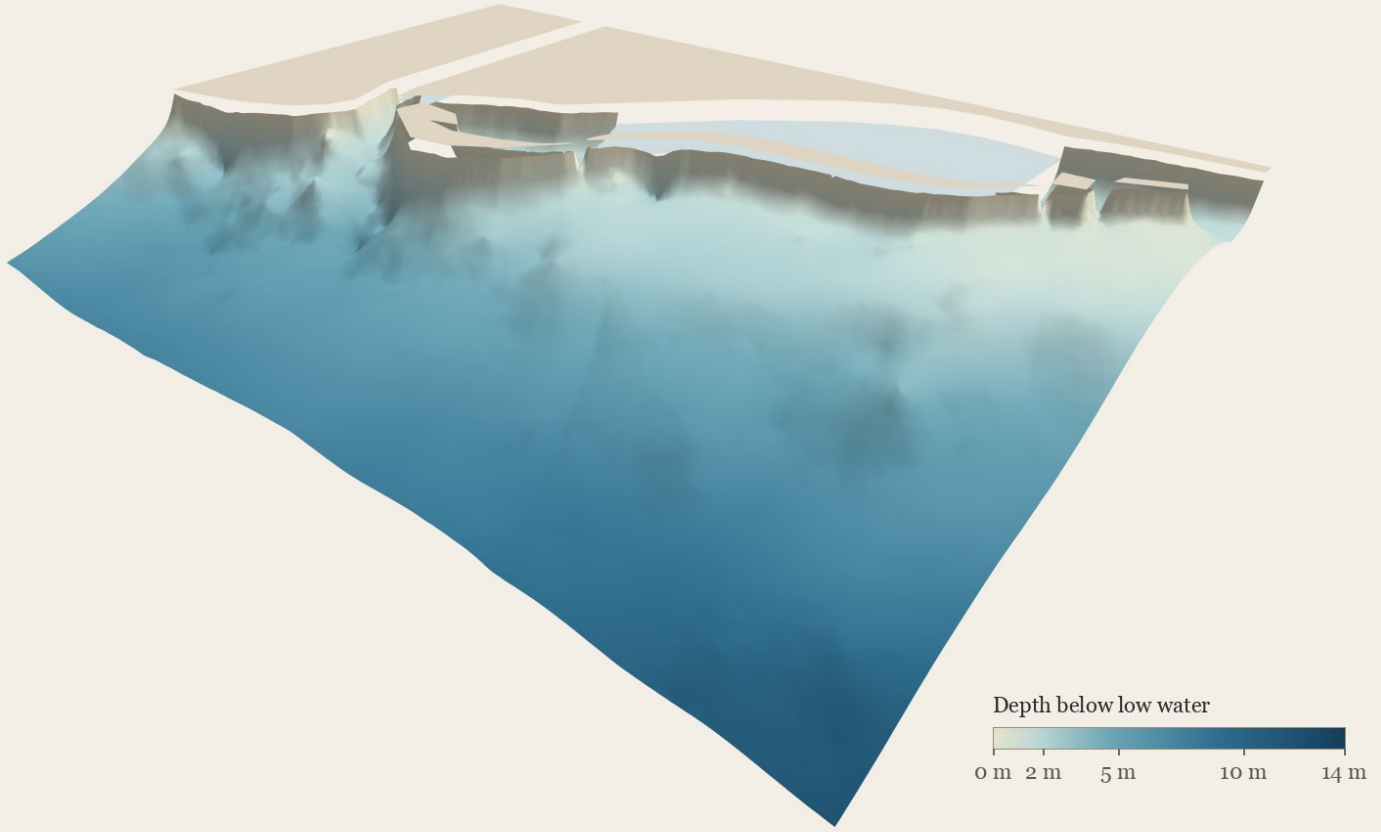
Rebuilt (130 soundings)



The seabed comes has an error of about 30cm in open water, 40cm along the shore, and ~1.2m in the inlet

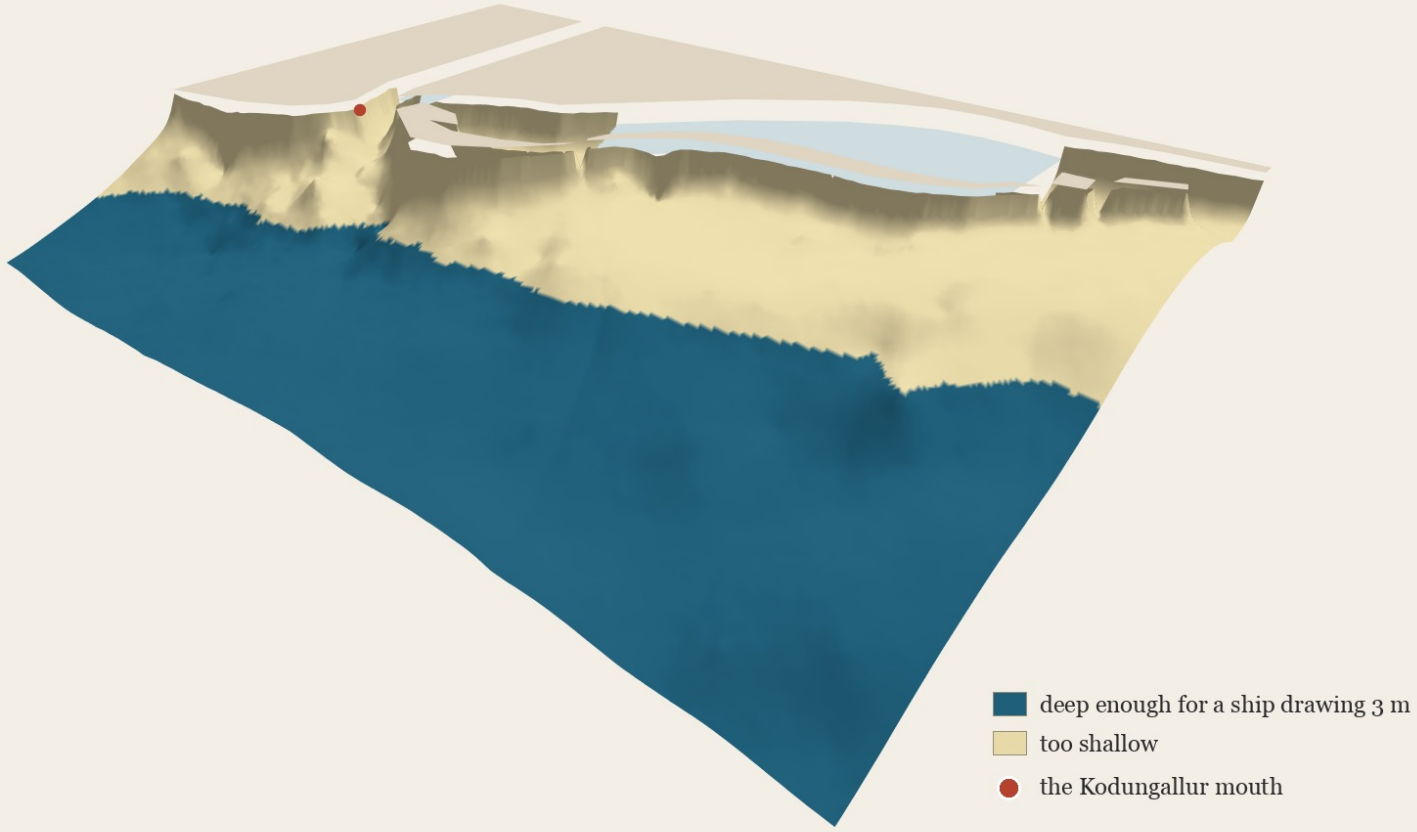
San Luis Pass, Texas: NOAA Continuously Updated DEM (CUDEM); survey simulated at the 1853 Kodungallur density

The coastal depths can be used to understand where ships can approach the shore



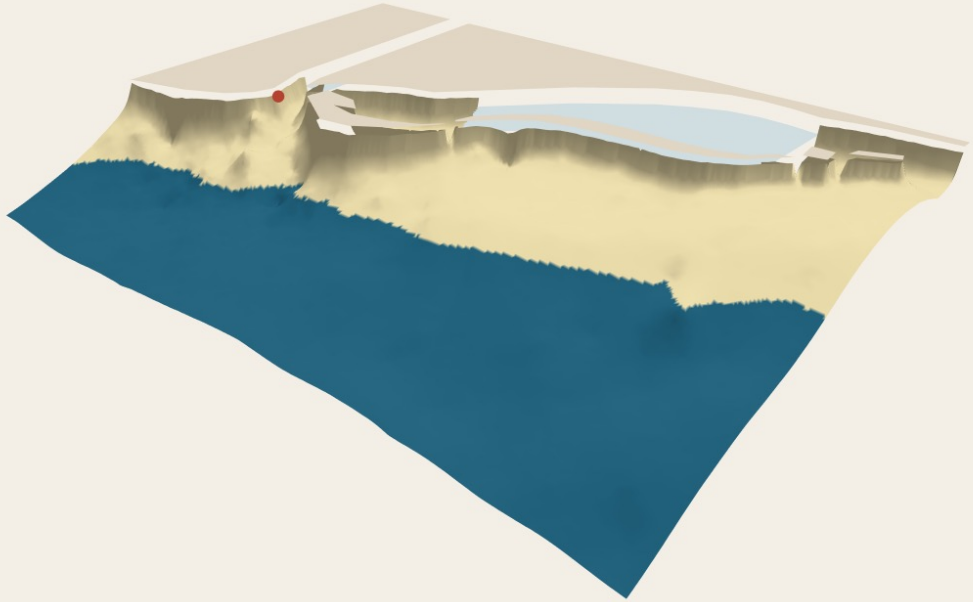
Soundings from Admiralty Chart 749; seabed modelled with a Gaussian process (R-INLA)

A loaded ship could not come in near the coast: water deep enough to accommodate it starts 1.3 km offshore



Soundings from Admiralty Chart 749; seabed modelled with a Gaussian process (R-INLA)

Pliny mentions this fact when talking about Muziris



praeterea longe a terra abest navium statio, lintribusque adferuntur onera et egeruntur.

"the roadstead for shipping is a long way from the land, and cargoes have to be brought in and carried out in boats."

Pliny, Natural History 6.104, on Muziris
Latin: Mayhoff 1906; translation: H. Rackham

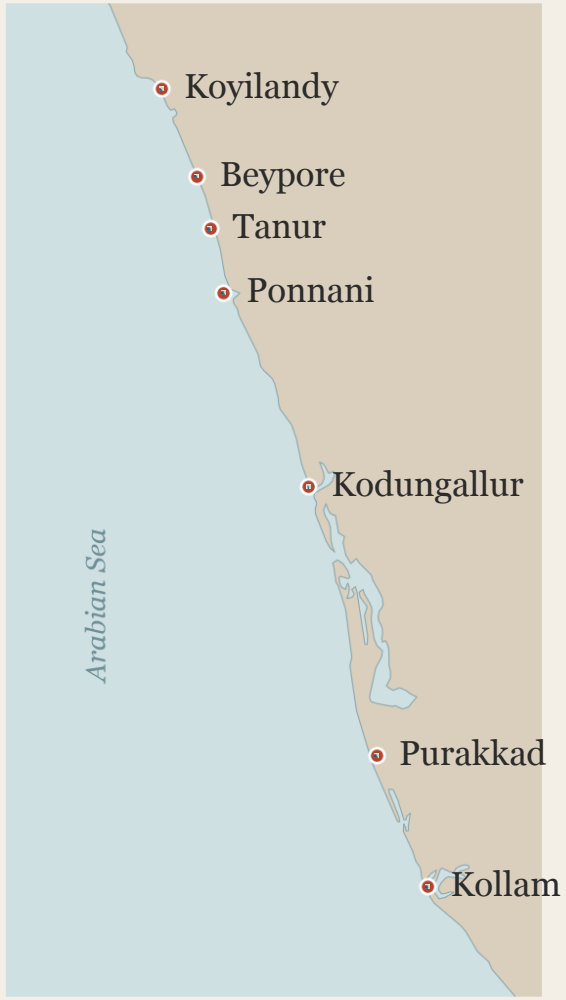
- deep enough for a 3 m draught (≥ 3.5 m, 90% sure)
- too shallow
- the Kodungallur mouth

Pliny, Natural History 6.104: Latin, ed. Mayhoff (Teubner, 1906); translation H. Rackham (Loeb). Soundings from Admiralty Chart 749; seabed modelled with a Gaussian process (R-INLA)

A loaded ship could not come in near the coast: water deep enough to accommodate it starts 1.3 km offshore

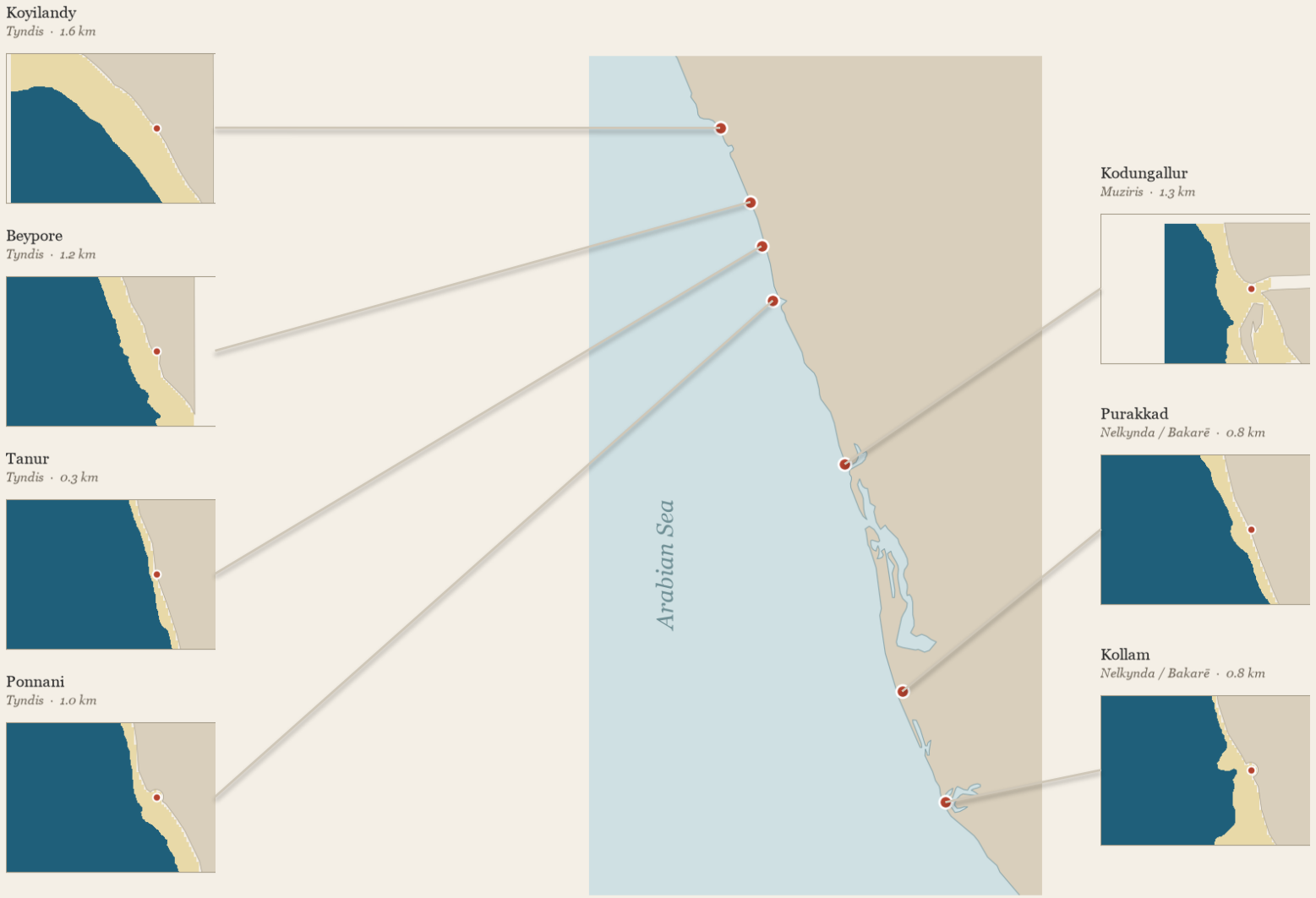
Soundings from Admiralty Chart 749; seabed modelled with a Gaussian process (R-INLA)

Similar analysis is run for the other possible port locations along the Malabar coat



Admiralty Charts 749, 750 (Selby, 1850–53) and 64 (1887); coast: Natural Earth; seabeds modelled by the author

Every candidate didn’t have sufficient coastal depth (in 1850) to enable Roman ships docking at the harbor



Admiralty Charts 749, 750 (Selby, 1850–53) and 64 (1887); coast: Natural Earth; seabeds modelled by the author

Part II

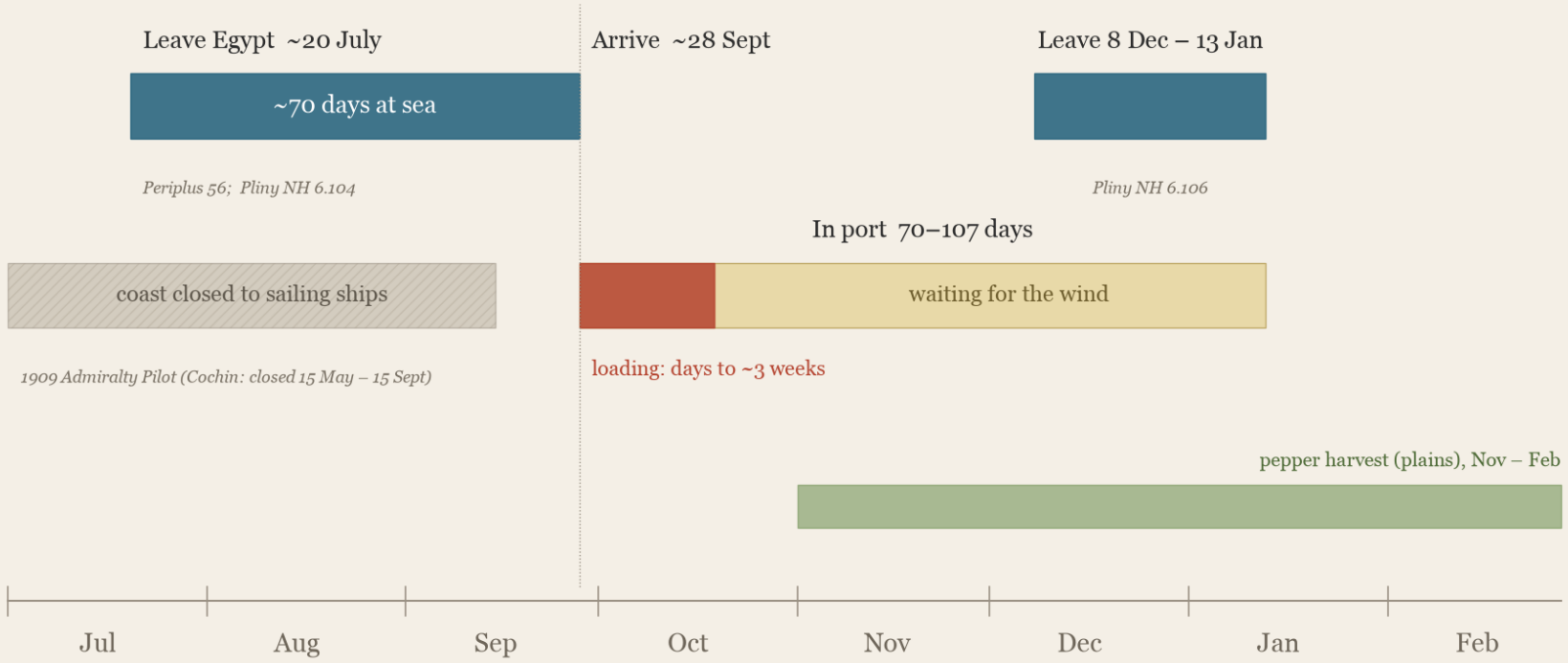
How did the ports work, and what came after Rome?

Chart: Admiralty Chart 750, off Porakad (Purakkad), Selby 1850–52, published 1857 (Wikimedia Commons, public domain)

The turnaround

How the port worked

Ships often lay offshore for months waiting for the monsoon wind even though loading took only weeks



Periplus 56; Pliny, Natural History 6.104, 6.106; Admiralty West Coast of India Pilot (1909); loading time: the author's estimate

The Pattanam wharf shows evidence for a dugout canoe and wheat & grain pips, supporting the coastal depth evidence

The dugout canoe in the waterlogged wharf deposit

The brick lining of the wharf

Wheat & grape pips and imported wheat was also found at the wharf.

"grain in sufficient amount for those involved with shipping, because the merchants do not use it" (Periplus 56)

Cherian and Menon (eds), Unearthing Pattanam (National Museum, New Delhi, 2014), pp. 120–21. Periplus 56, trans. Casson (1989: 85)

While we have evidence that Indians lived in Roman ports, there is no definitive evidence that Romans lived in the ports along the Malabar coast

Indians had household objects in Red Sea ports

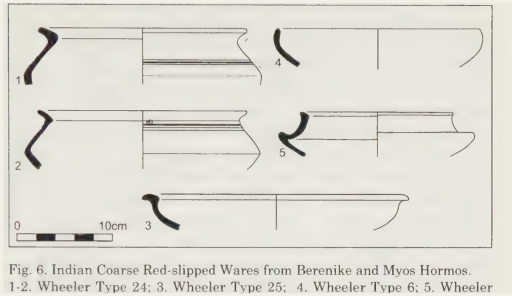


Fig. 6. Indian Coarse Red-slipped Wares from Berenike and Myos Hormos. 1-2. Wheeler Type 24; 3. Wheeler Type 25; 4. Wheeler Type 6; 5. Wheeler

Indian cooking pots, Berenike and Myos Hormos

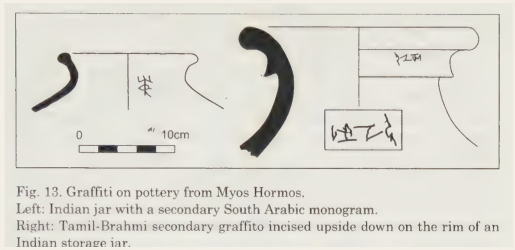


Fig. 13. Graffiti on pottery from Myos Hormos.
Left: Indian jar with a secondary South Arabic monogram.
Right: Tamil-Brahmi secondary graffito incised upside down on the rim of an Indian storage jar.

Tamil-Brahmi names scratched on jars, Myos Hormos

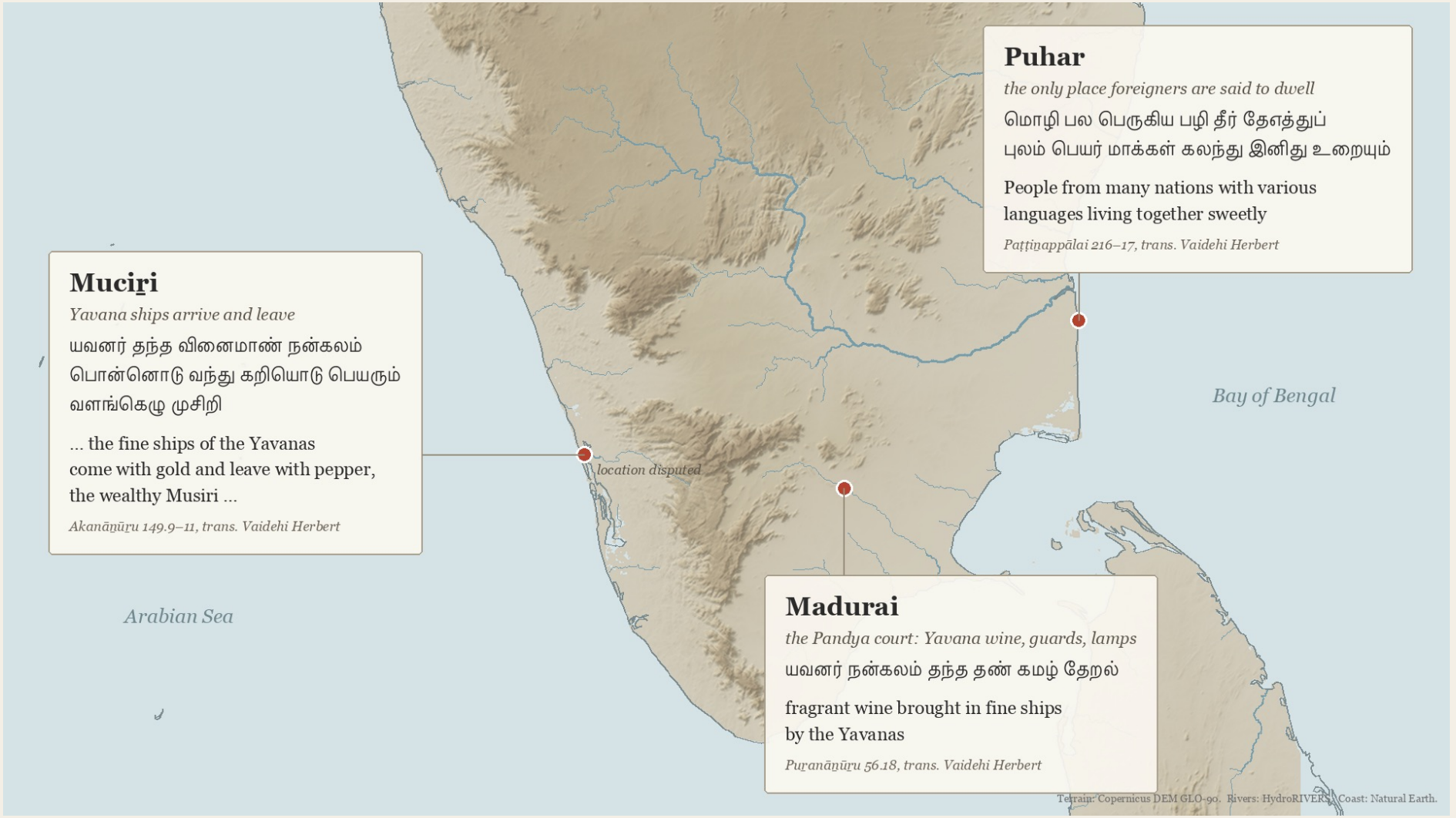
The only Roman objects at Pattanam are trade goods

Dressel 2-4 wine amphora from Campania

Stamped Italian terra sigillata

Tomber, Indo-Roman Trade (2008), figs 6 and 13; Shajan, Tomber, Selvakumar and Cherian, JRA 17 (2004), fig. 5; Cherian and Menon (eds), Unearthing Pattanam (National Museum, New Delhi, 2014)

Exant tamil poetry mention foreigners living at Puhar, but none at Muziris

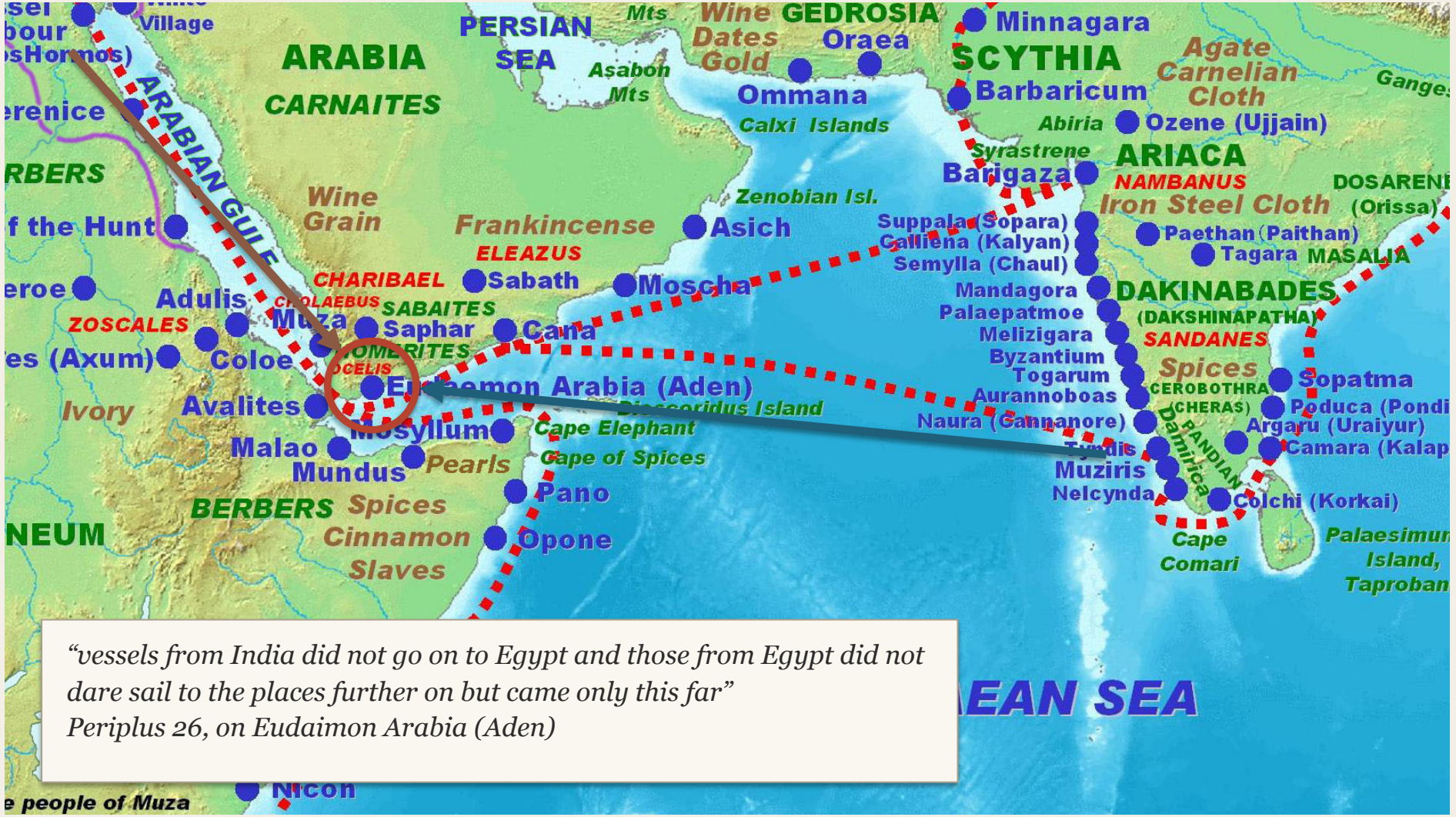


Translations: Vaidehi Herbert (sangamtranslationsbyvaidehi.com). Terrain: Copernicus DEM; rivers: HydroRIVERS; coast: Natural Earth

The crossing

When south India began to cross the Arabian Sea

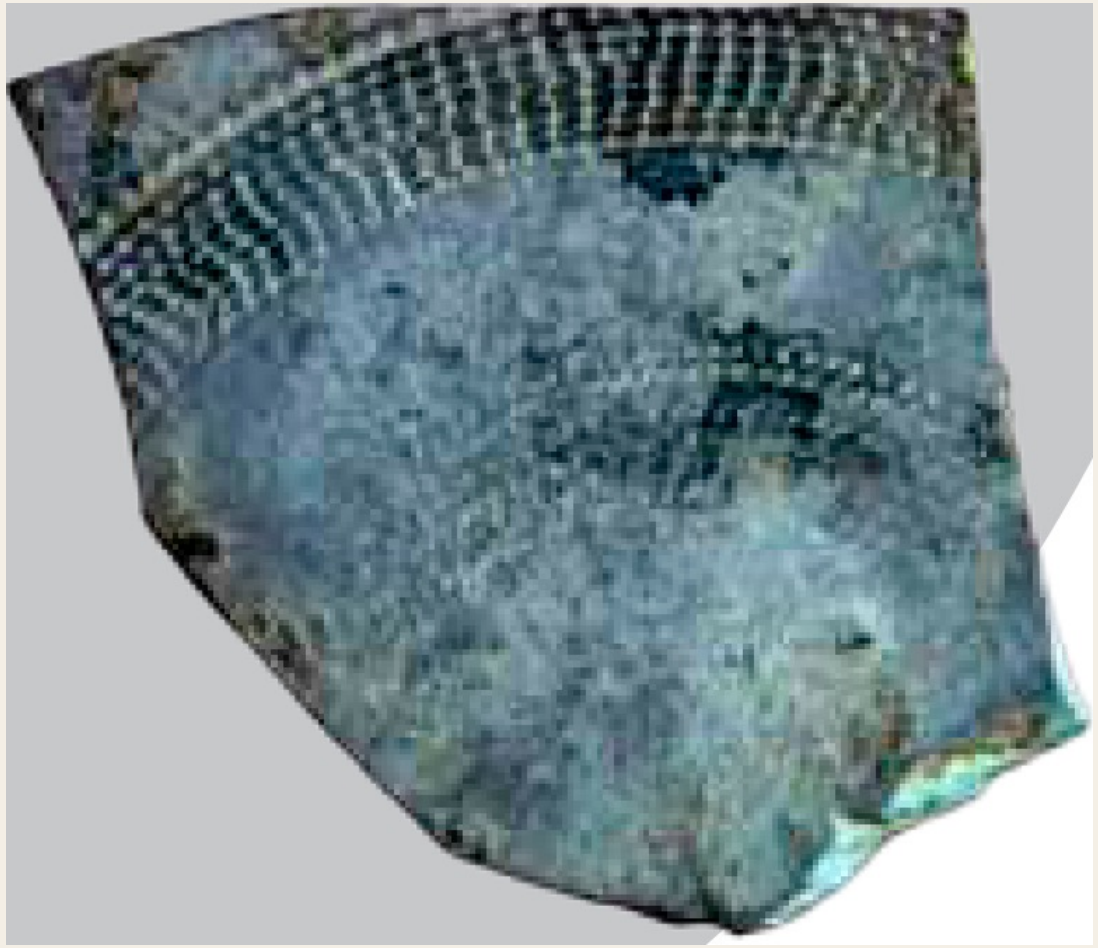
Before the Periplus, Indian and Egyptian ships met at Aden, indicating that Indian goods reached Rome by relay



"vessels from India did not go on to Egypt and those from Egypt did not dare sail to the places further on but came only this far"
Periplus 26, on Eudaimon Arabia (Aden)

Periplus 26, trans. Casson (1989: 65). Map: PHGCOM, 'Map of the Periplus of the Erythraean Sea', Wikimedia Commons, CC BY-SA 4.0; arrows added by the author

Wheeler misread Indian pottery at Arikamedu to argue for a Roman trading settlement in India



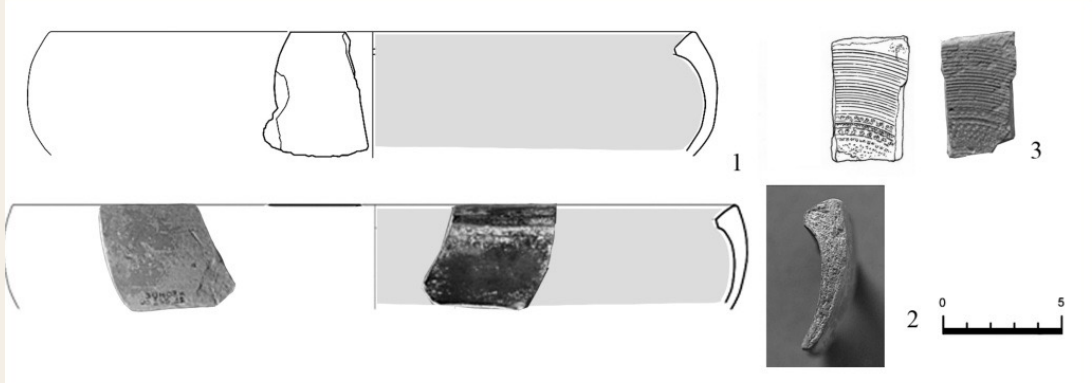
Rouletted Ware, Pattanam

1 Misread as Roman

- Initially mistaken for Roman pottery because of its fine finish and rouletted decoration, which resembled Greco-Roman ceramics.
- Its shape was later argued to be Indian (Begley), which was then confirmed by X-ray analysis (Gogte).

Pattanam sherd: Cherian and Menon (eds), Unearthing Pattanam (2014), p. 50. Tomber, Indo-Roman Trade (2008: 44–45); Pavan and Schenk, AAE 23 (2012)

Later scholars misdated similar pottery in Arabia to argue that south Indian merchants crossed before the Romans



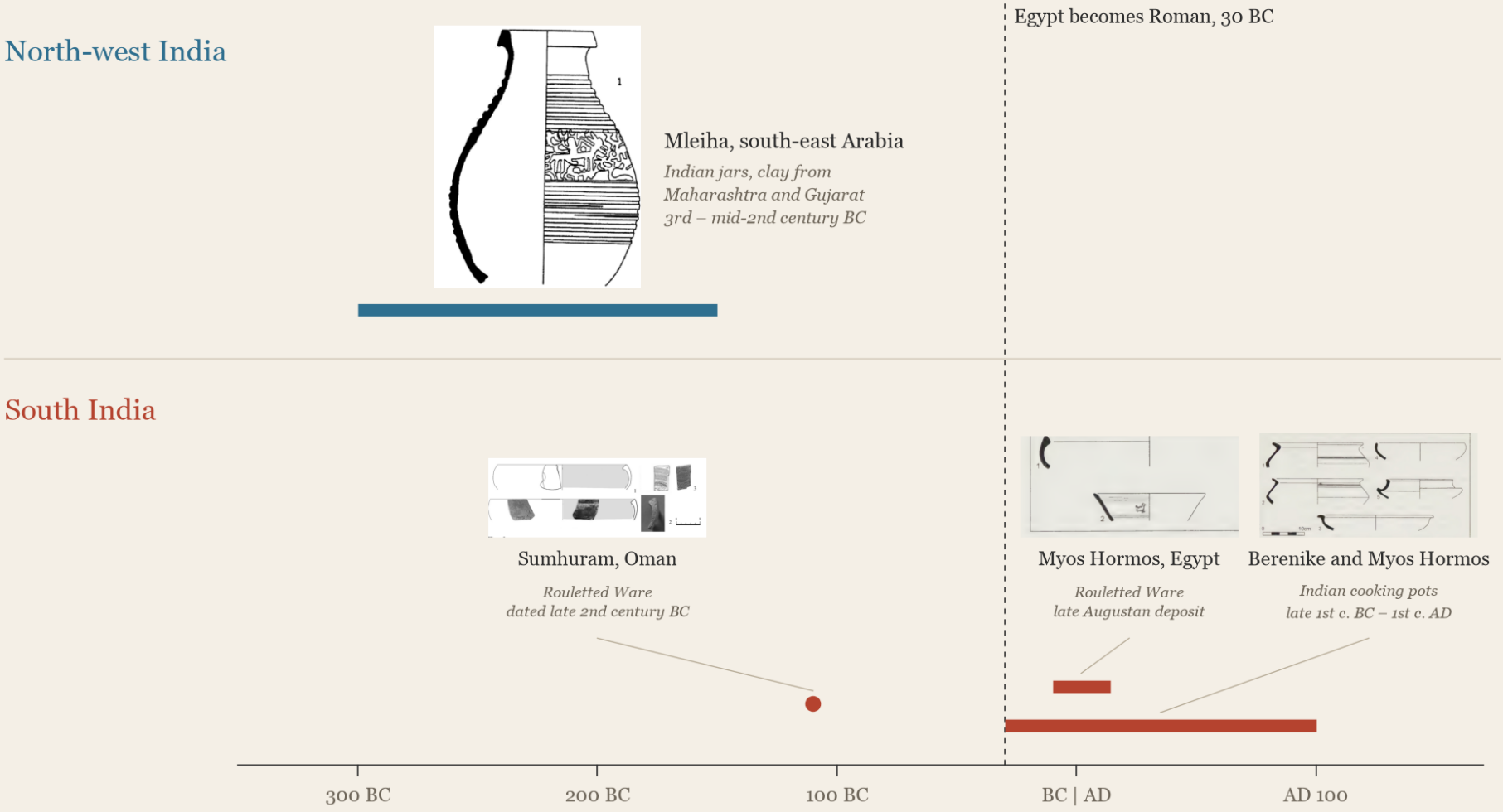
Rouletted Ware, Sumhuram (Oman)

2 Misdated as early

- Similar ware found at Sumhuram in Oman was dated to the late 2nd century BC by matching it to finds in Sri Lanka, and was taken as evidence that south India traded across the Arabian Sea before Rome.
- Yet at Sumhuram itself, the sherds were found near the surface, in later layers, or alongside Roman amphorae of the late 1st century BC.
- Fine pottery like this was often kept as an heirloom for generations, so these pieces could have arrived long after they were made, after Roman contact had begun.

Sumhuram sherd: Pavan and Schenk, AAE 23 (2012), fig. 3. Tomber, Indo-Roman Trade (2008: 44–45)

While north-west Indian pottery reached Arabia before Rome, nothing from south India is securely dated before the Roman trade

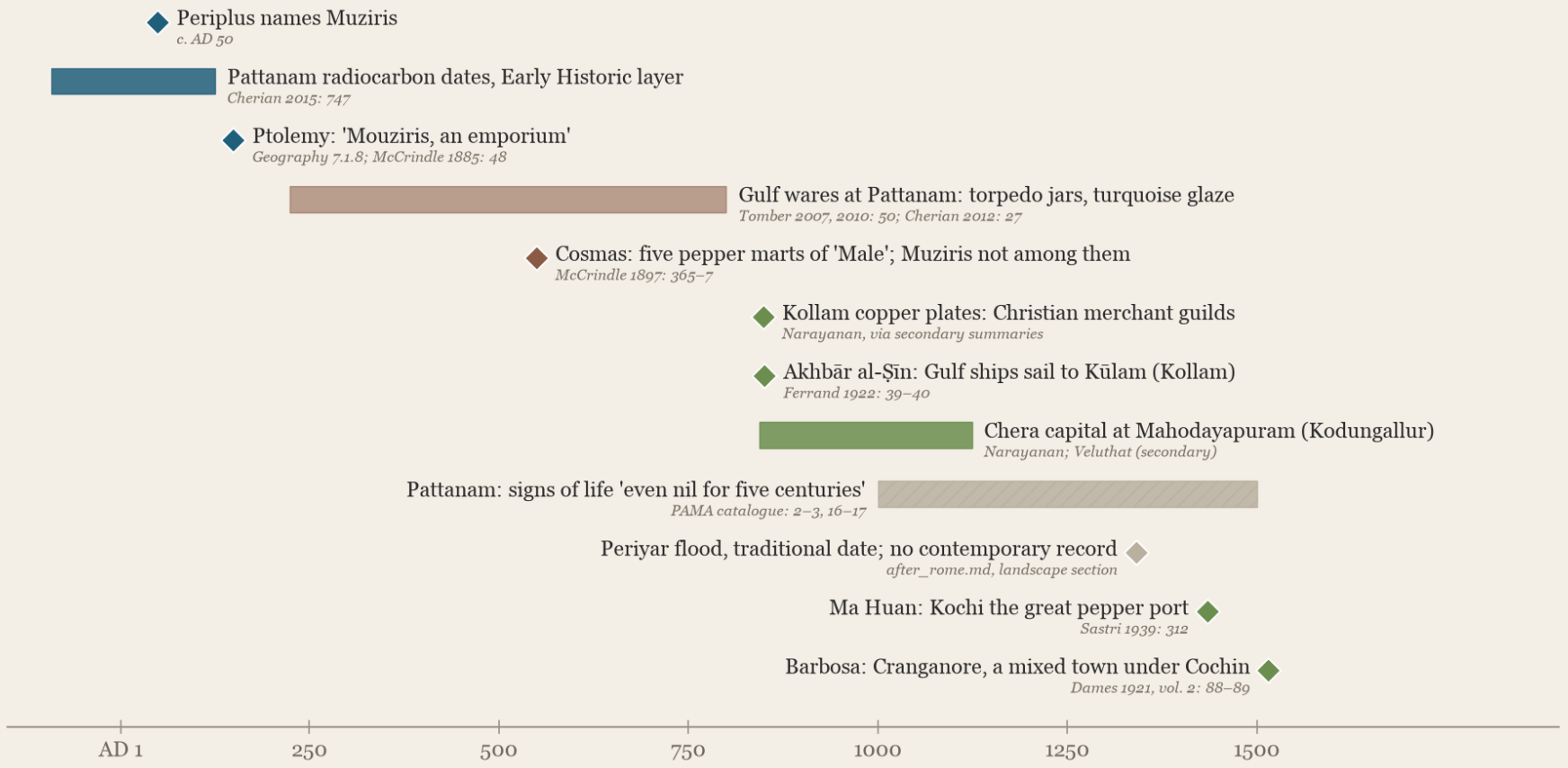


Reddy 2013: 32, fig. 213 (after Mouton 2008); Pavan and Schenk 2012, fig. 3; Tomber 2008, figs 5–6, pp. 74–75; Shoebridge 2017: 112–13

After Rome

What became of the port when Rome's trade faded

When Roman trade faded, Pattanam turned to the Gulf, and the coast's main port moved to Kollam, Kodungallur and then Kochi



Each item's source is printed beneath it; the copper plates and the Chera dates rest on secondary summaries

After Rome, Pattanam's foreign pottery comes mostly from the Gulf

Torpedo jar sherd with bitumen lining

Turquoise-glazed ring base

Cherian and Menon (eds), Unearthing Pattanam (National Museum, New Delhi, 2014)

Painigunum
Metrudah
Kranganur
Pallipur R.
Maniput
Pallipur

Questions